🔐 Introduction to Network Security Challenges

📌 What is Network Security? (Definition)

- **Network Security** means protecting a computer network from:
  - Unauthorized access
  - Data theft
  - Misuse or attacks
- It ensures that:
  - Data is **safe**
  - Users are **authorized**
  - Network works **properly**

⚠️ Types of Network Security Challenges

1. 🕵️ Unauthorized Access : When someone enters the network without permission Example: Hacking WiFi password Accessing private data
2. 🦠 Malware Attacks Harmful software like: Virus Worm Trojan Effects: Damage system Steal data
3. 🎣 Phishing Attacks Fake emails or messages to trick users Goal: Steal passwords Get bank details
4. 💣 Denial of Service (DoS) Attack Attackers send too much traffic Network becomes: Slow Crashed
5. 🔓 Weak Passwords Easy passwords like: 123456 password Makes hacking very easy

Made with GAMMA

Essential Security Measures: Firewalls and VPNs

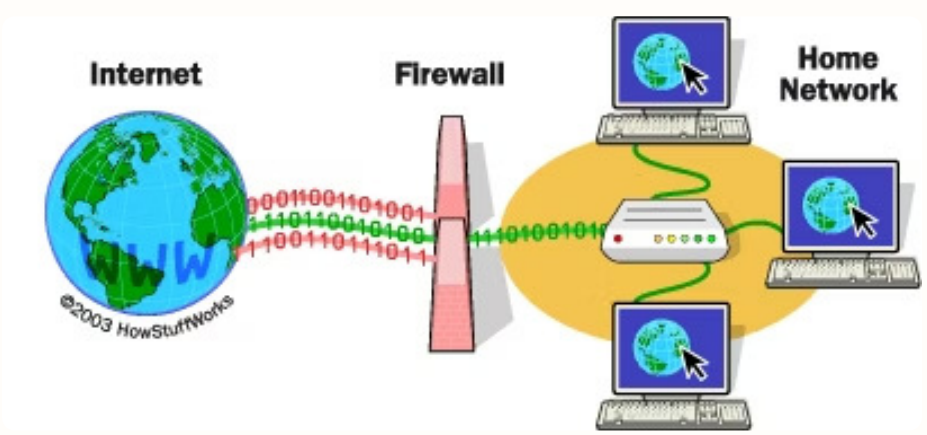
Two fundamental tools form the backbone of network security: firewalls and Virtual Private Networks (VPNs). These technologies work together to create secure boundaries and encrypted pathways for data transmission. Understanding how they function will help you make informed decisions about protecting your digital presence.

Firewall Types

- **Packet-filtering firewalls:** Examine individual data packets
- **Stateful firewalls:** Track active connections for context
- **Proxy firewalls:** Act as intermediaries for requests
- **Next-generation firewalls:** Include additional security features

VPN Protocols

- **OpenVPN:** Open-source with strong security
- **IPSec:** Enterprise-level encryption standards
- **WireGuard:** Modern, faster, and simpler protocol
- **SSL/TLS:** Common for web-based secure connections



Made with GAMMA